

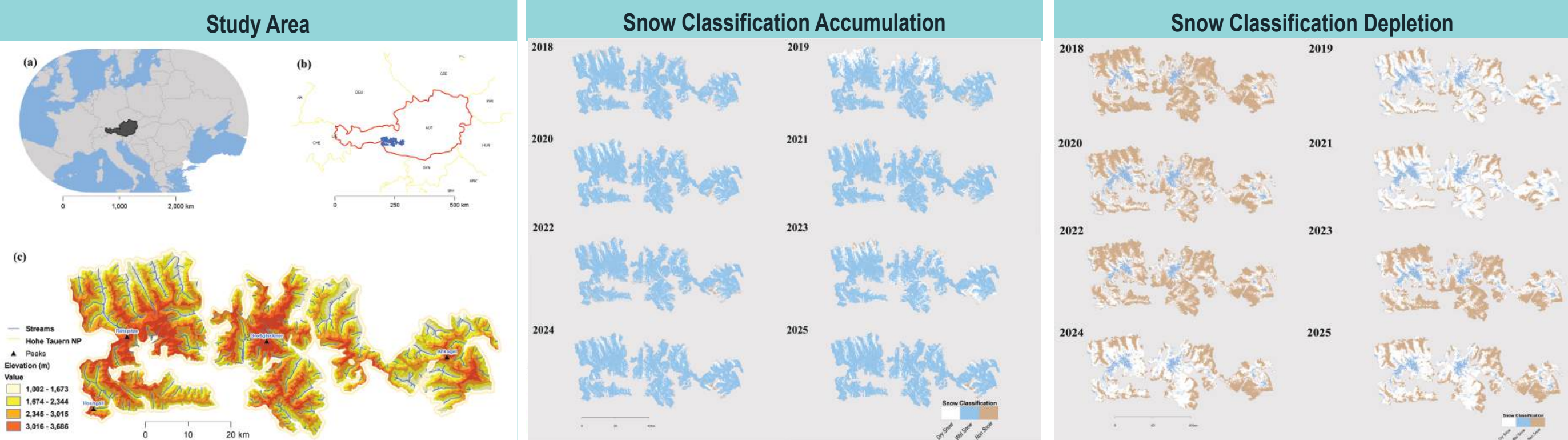
INTEGRATION OF SAR AND OPTICAL SATELLITE DATA FOR DELINEATION OF WET AND DRY SNOW

HOHE TAUERN NATIONAL PARK, AUSTRIA

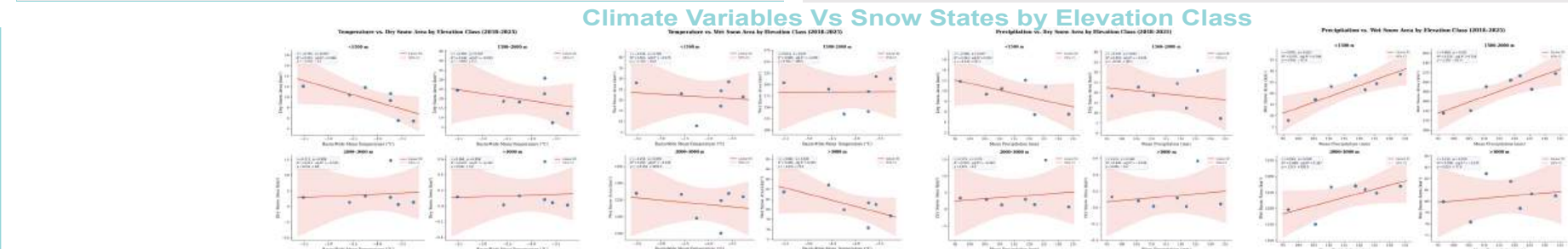
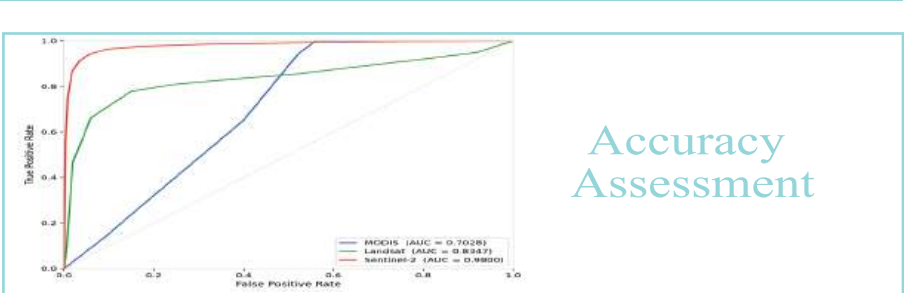
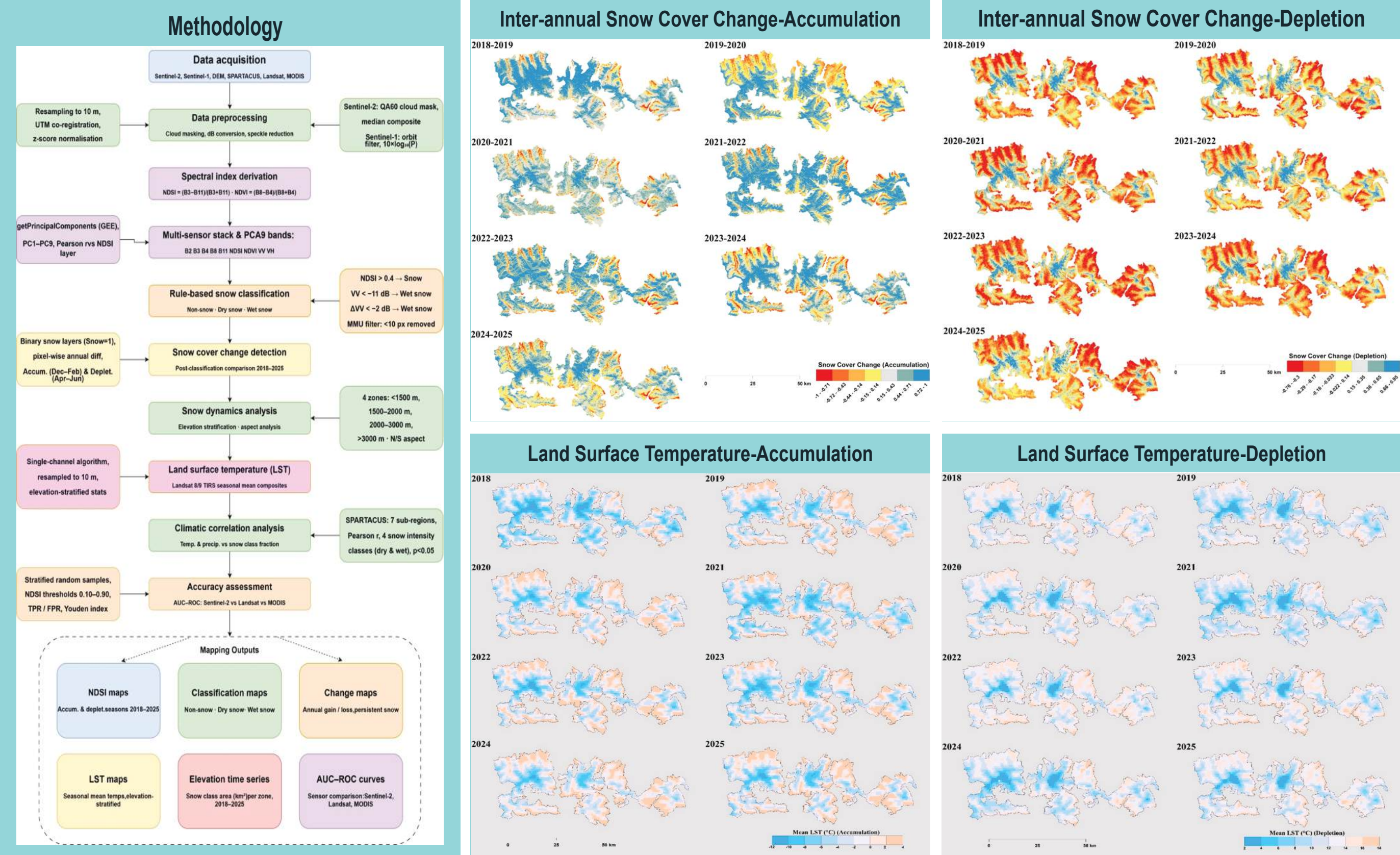
Diploma Thesis

Goal: This study integrates Sentinel-1 SAR and Sentinel-2 optical imagery to detect snow cover and to distinguish wet from dry snow across Hohe Tauern National Park, Austria (2018-2025). Three specific objectives are pursued:

1. Fuse Sentinel-1 and Sentinel-2 via PCA for improved snow detection.
2. Classify dry and wet snow via NDS + SAR backscatter rules.
3. Link snow states to temperature, precipitation and snowmelt dynamics.



This work delivers reproducible, GEE-based multi-sensor framework for delineating dry and wet snow in alpine terrain. It quantifies how warming is shifting the dry-to-wet snow boundary upward, and offers an operational baseline for cryosphere monitoring, hydrological forecasting and climate-adaptation planning in Alpine protected areas.



S1/S2 Fusion Excellence: Integrated Sentinel-1 SAR and Sentinel-2 Optical data in Google Earth Engine to achieve 100% cloud-free snow monitoring (2018–2025).

High Precision Mapping: Sentinel-2 NDSI achieved superior classification accuracy (AUC = 0.98) compared to Landsat (0.83) and MODIS (0.70).

Wet/Dry Delineation: Successfully distinguished Non-Snow, Dry Snow, and Wet Snow states; SAR backscatter change detection reliably identified liquid water content.

Alpine Warming Evidence: Observed an accelerating upward shift of the seasonal snowline and a decrease in dry snow extent even above 3,000 m (loss of ~1.1 km² in high-altitude zones).

Scalable Framework: Developed a reproducible, GEE-based multi-sensor workflow for operational alpine cryosphere monitoring.